

Haramrit Singh Khurana

Dallas-Fort Worth, TX — haramrit09k@gmail.com — linkedin.com/in/haramrit09k — github.com/haramrit09k

Summary

Software Engineer with 4+ years building and owning backend/full-stack systems in a regulated enterprise environment (Java/Spring Boot, Angular, OpenShift/Kubernetes), plus hands-on LLM/agent tooling work and a published distributed-ML-training research paper.

Experience

Software Engineer, Citi

DFW, TX

2022 – Present

- Architected and built the ServiceNow CHG ticket creation automation for DBaaS self-service operations, eliminating manual Ops-team involvement and enabling fully automated, instant ticket creation for 300–2,000 CHG tickets/month (peak during major MongoDB releases)
- Owned the Angular 19 upgrade end-to-end for the MaaS application and the shared DBaaS component library, coordinating breaking-change fixes across 4 consumer teams (MSaaS, OraaS, PgaaS, CaaS); removed 10 unused components/3 dead services, cutting build time 4m48s → 1m34s (67% faster) and `ng serve` startup 100s → 35-40s
- Designed and built a dynamic, self-service approval-group configuration system in Spring Boot, replacing hardcoded approval chains after a company-wide reorg; eliminated approval-related code changes entirely and saved 2–3 developer hours/sprint with a fully backward-compatible rollout
- Migrated the daily CAI (Citi Application Inventory) data ingestion pipeline from a raw SQL cronjob to a GraphQL-based sync in an existing Spring Boot worker microservice, processing 150K rows × 130 columns daily to power auth/approval routing platform-wide
- Designed a bulk VM/filesystem metadata extraction workflow (GraphQL + JDBC, 30K+ rows); identified a scalability risk in an agreed single-table design, benchmarked alternatives, and led adoption of a normalized schema cutting bulk insert time 50% (10–12s → 5–6s) under a compressed timeline
- Independently root-caused a production outage (portal-wide request timeouts) by tracing frontend, OpenShift logs, and an inefficient 7–8 join DB view; shipped a simplified replacement view as a safe hotfix while senior engineers were unavailable, later adopted permanently
- Implemented HTTP Digest Authentication (RFC 7616) from scratch as a custom Feign interceptor to integrate MongoDB Ops Manager metrics; passed security review and was reused by another team, saving them 1 week of effort
- Solo-built a Playwright E2E test automation framework running unattended via CI on OpenShift with self-validating assertions, cutting manual regression testing 60% while active; introduced Husky pre-commit hooks and cleaner PR standards, reducing PR rework from 3–4/sprint to 1–2/sprint

Software Engineer, Optimal SatCom Inc.

Herndon, VA

Jun 2021 – Oct 2022

- Contributed to migrating legacy firm-wide applications to a modern stack (DevExpress React, .NET Core, MS SQL)
- Designed and implemented key components in C#/.NET Core, owning ongoing enhancements, bug fixes, and resolution of silent console crashes and security-deprecation issues
- Revamped database architecture for the company's Enterprise Capacity Management (ECM) SATCOM system – triggers, queries, stored procedures, upgrade scripts
- Reduced compiler warnings/errors by 70% through legacy code optimization, collaborating with QA/support on urgent bug fixes

Teaching Assistant, IT207: Applied IT Programming, George Mason University

Fairfax, VA

Jan 2020 – May 2021

- Lab Instructor/GTA covering HTML, CSS, PHP, MySQL; supported 60 students individually during office hours
- Created an assignment submission format standard adopted by other TAs across additional course sections

Publications

A Resource Effective Approach for Distributing Machine Learning Models over a Cluster – ICAC3 2019 (IEEE)

Paper — Demo

- Built the desktop application automating distribution of ML model training across a local network cluster; contributed to distribution-logic design
- Achieved a 10% reduction in model training time with no accuracy loss; 1st Runner-Up, DBIT Innovex capstone competition

Projects

LogScribe-MCP – Python MCP server exposing log-query and pattern-detection tools to LLM clients; open source on GitHub

Local Resume Optimization Pipeline – Offline pipeline using MCP tooling and Ollama-hosted models to optimize resume content against a target job description

Education

Executive M.S., IT Project Management – New England College (expected mid-2026)

B.E., Computer Engineering – University of Mumbai (DBIT Kurla), 2015 – 2019

Certifications

AWS Certified Solutions Architect – Associate

Skills

Java, Spring Boot, Angular, TypeScript, C#/.NET Core, Python, SQL, GraphQL, OpenShift, Kubernetes, Helm, Playwright, Git, CI/CD, MCP/agent tooling, LLM prompt engineering